

IMO Training – Number Theory

Miscellaneous Problems 2

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Notes

Unless otherwise specified, let p be a prime number.

Theorem. (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a, p) = 1, a^{p-1} \equiv 1 \pmod{p}$.

Definition. (Euler Totient Function) $\forall n \in \mathbb{N}, \varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m, n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$.

Theorem. (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a, n) = 1, a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem. (Wilson's Theorem) p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Definition. (Order and Primitive Roots) For $a \in \mathbb{Z}, n \in \mathbb{N}$ with $(a, n) = 1$, the *order of a modulo n* ($\text{ord}_n(a)$) is defined by the minimal $m \in \mathbb{N}$ s.t. $a^m \equiv 1 \pmod{n}$. Furthermore, if $\text{ord}_n(a) = \varphi(n)$, then a is a *primitive root modulo n* .

Definition. (Valuation) For $m \in \mathbb{N}$, the *m -adic valuation* of n is the $a \in \mathbb{N}$ s.t. $m^a \mid n$ and $m^{a+1} \nmid n$.

Exercises

1. Find all primes p, q, r such that $p^2 + q^3 = r^4$.
2. (Senior Hanoi Open 2006) Calculate the last 3 digits of $2005^{11} + 2005^{12} + \dots + 2005^{2006}$.
3. (PUMAC) Calculate the last 3 digits of $2008^{2007^{2006 \dots 2^1}}$.
4. (Canada 2003) Calculate the last 3 digits of $2003^{2002^{2001}}$.
5. (ISL 2005 N6) Let $a, b \in \mathbb{N}$ such that $a^n + n \mid b^n + n$ for all $n \in \mathbb{N}$. Prove that $a = b$.
6. (USA TST 2003) Find all primes p, q, r such that $p \mid q^r + 1, q \mid r^p + 1, r \mid p^q + 1$.
7. Prove that for $n > 1$, $n \nmid 2^{n-1} + 1$.
8. (China 2009) Find all primes p, q such that $pq \mid 5^p + 5^q$.